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Apparatus and method of substrate edge cleaning and substrate carrier head gap cleaning

Abstract

The present disclosure relates to load cups that include an annular substrate station configured to receive a substrate. The annular substrate station surrounds a nebulizer located within the load cup. The nebulizer includes a set of energized fluid nozzles disposed on an upper surface of the nebulizer adjacent to an interface between the annular substrate station and the nebulizer. The set of energized fluid nozzles are configured to release energized fluid at an upward angle relative to the upper surface.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7044832	12/2005	Yilmaz	451/5	B24B 37/345
2006/0255016	12/2005	Svirchevski et al.	N/A	N/A
2010/0291841	12/2009	Sung	451/177	B24B 53/017
2016/0052104	12/2015	Ishibashi	156/345.14	B24B 37/10
2019/0157071	12/2018	Peng	N/A	H01L 21/30625
2020/0234995	12/2019	Rangarajan	N/A	H01L 21/67051
2020/0331113	12/2019	Soundararajan et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111469046	12/2019	CN	N/A
2016-043442	12/2015	JP	N/A
2018046260	12/2017	JP	N/A
2018-0133512	12/2017	JP	N/A
10-2016-0023568	12/2015	KR	N/A
10-2018-0137136	12/2017	KR	N/A
10-2019-0018728	12/2018	KR	N/A
2008079449	12/2007	WO	N/A
2020/150072	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report/ Written Opinion issued to PCT/US2021/053907 on Jan. 10, 2022. cited by applicant
Search Report for Taiwan Application No. 110140518 dated Feb. 3, 2023. cited by applicant
Office Action for Taiwan Application No. 110140518 dated Mar. 8, 2023. cited by applicant
JP Office Action dated Oct. 10, 2023 for Application No. 2022-544300. cited by applicant
Notice of Preliminary Rejection Dated May 28, 2024, re: Korean patent application No. 10-2022-

7026263. cited by applicant
Search Report issued to Taiwan Application No. 112140745 on Feb. 24, 2025 in 8 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/091,105, filed Nov. 6, 2020, which is herein incorporated by reference in its entirety.

BACKGROUND
Field
(1) Embodiments of the present disclosure generally relate to substrate processing, and more specifically to substrate processing tools and methods thereof.
Description of the Related Art
(2) An integrated circuit is typically formed on a substrate by the sequential deposition of conductive, semiconductive or insulative layers on a silicon substrate. Fabrication includes depositing a filler layer over a non-planar surface, and planarizing the filler layer until the non-planar surface is exposed. A conductive filler layer can be deposited on a patterned insulative layer to fill the trenches or holes in the insulative layer. The filler layer is then polished until the raised pattern of the insulative layer is exposed. After planarization, the portions of the conductive layer remaining between the raised pattern of the insulative layer form vias, plugs and lines that provide conductive paths between thin film circuits on the substrate. In addition, planarization may be needed to planarize a dielectric layer at the substrate surface for photolithography.
(3) Chemical mechanical polishing (CMP) is one accepted method of planarization. This planarization method includes mounting the substrate on a carrier head or polishing head of a CMP apparatus. The exposed surface of the substrate is placed against a rotating polishing disk pad or belt pad. The carrier head provides a controllable load on the substrate to urge the device side of the substrate against the polishing pad. A polishing slurry, including at least one chemically-reactive agent, and abrasive particles if a standard pad is used, is supplied to the surface of the polishing pad.
(4) The substrate is typically retained below the carrier head against a membrane within a retaining ring. Moreover a gap is present between an outer edge of the substrate and an inner perimeter of the retaining ring when the substrate is in the carrier head. In addition, a gap is present between an outer edge of the membrane and an inner perimeter of the retaining ring. These gaps and other areas proximate to the outer edge of the substrate can accumulate polishing slurry and organic residues during processing. These residues can remain on the substrate edge and/or dislodge during processing and cause defects to the substrate and affect the efficiency of the CMP apparatus. Thus, there is a need for a method of removing residue from the substrate edge and removing residue from the gaps in the carrier head surrounding the substrate. There is also a need for an apparatus for removing the residue from the substrate edge before the substrate is transferred off of the carrier head and for removing the residue from the gaps surrounding the membrane of the carrier head.
SUMMARY

(5) In one embodiment, a load cup is provided having an annular substrate station configured to receive a substrate and a nebulizer located within the load cup and surrounded by the annular substrate station. The nebulizer has a set of energized fluid nozzles disposed on an upper surface of the nebulizer adjacent to an interface between the annular substrate station and the nebulizer. The set of energized fluid nozzles are configured to release energized fluid at an upward angle relative to the upper surface.
(6) In another embodiment, a method of cleaning a chemical mechanical polishing system is provided including directing energized fluid from a set of energized fluid nozzles of a load cup at an edge of a substrate disposed in a carrier head. The carrier head has a retaining ring to retain the substrate below a membrane of the carrier head. The method includes unloading the substrate from the carrier head and directing the energized fluid from the set of energized fluid nozzles to a gap formed between an outer edge of the membrane and an inner perimeter of the retaining ring.
(7) In another embodiment, a chemical mechanical polishing system includes a carrier head having a retaining ring to retain a substrate below a membrane of the carrier head the chemical mechanical polishing system includes a load cup having a set of energized fluid nozzles disposed on an upper surface of an outer portion of the load cup. The set of energized fluid nozzles are configured to direct energized fluid to a gap between an outer edge of the membrane and an inner perimeter of the retaining ring.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, and the disclosure may admit to other equally effective embodiments.
(2) FIG. **1** depicts a top, plan view of a chemical mechanical polishing (CMP) system according to an embodiment.
(3) FIG. **2**A depicts a partial side view of a polishing apparatus according to an embodiment.
(4) FIG. **2**B depicts a bottom view of a polishing apparatus according to an embodiment.
(5) FIG. **3**A depicts a top, plan view of a load cup according to an embodiment.
(6) FIG. **3**B depicts a schematic side view of a spray pattern for a nozzle according to an embodiment.
(7) FIG. **4** depicts a flow diagram of a method for processing a substrate according to an embodiment.
(8) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.
DETAILED DESCRIPTION
(9) The present disclosure relates to load cups that are configured with energized fluid nozzles which expel tunable energized fluid jets to clean the edges of a substrate disposed in a carrier head before the substrate is unloaded from the carrier head without overheating the substrate. The energized fluid jets expelled from the energized fluid nozzles have characteristics that are favorable for penetrating and effectively cleaning narrow gaps in the carrier head between a hydrophobic membrane and an inner perimeter of a retaining ring of the carrier head. According to one or more embodiments of the disclosure, it has been discovered that certain properties of the energized fluid

jets, such as one or more of a flat fan shape, high pressures, high temperatures, gas phase stream, solid, meltable particles for bombardment, sonic wave generation, and combination(s) thereof may be advantageously used for enhanced and quick cleaning. It has also been found that each of the energized fluid nozzles can be used in combination with respective fluid spray nozzles (e.g., deionized water spray nozzles) for improved control such as temperature control.
(10) FIG. **1** depicts a top, plan view of a chemical mechanical polishing (CMP) system **100** according to an embodiment disclosed herein. Although a CMP system is illustrated in FIG. **1** and described herein, the concepts disclosed herein may be applied to other substrate processing devices. The CMP system **100** includes a polishing section **102** and a cleaning and drying section **104** that process (e.g., wash and/or polish) substrates **108**. The CMP system **100** includes other sections that perform other processes on the substrates **108**. As used herein, substrates include articles used to make electronic devices or circuit components. Substrates include semiconductor substrates such as silicon-containing substrates, patterned or unpatterned substrates, glass plates, masks, and the like. A pass-through **110** is an opening between the polishing section **102** and the cleaning and drying section **104** that accommodates the transfer of the substrates **108**.
(11) The polishing section **102** includes one or more polishing stations **114**, such as individual polishing stations **114**A-**114**D. Each of the polishing stations **114** include a polishing pad, such as individual polishing pads **116**A-**116**D. The polishing pads rotate against surfaces of the substrates **108** to perform various polishing processes. One or more slurries (not shown) are applied between the substrate **108** and the polishing pad **116**A-**116**D to process the substrate.
(12) The polishing section **102** includes a plurality of carrier heads **120** that maintain the substrates **108** against the polishing pads **116**A-**116**D during polishing. Each of the polishing stations **114**A-**114**D may include a single head, such as individual carrier heads **120**A-**120**D. The carrier heads **120**A-**120**D secure the substrates **108** therein as the carrier heads **120**A-**120**D are transported to and from the polishing stations **114**A-**114**D. For example, the carrier heads **120**A-**120**D secure the substrates **108** therein as the carrier heads **120**A-**120**D are transported between load cups **124** (e.g., individual load cups **124**A, **124**B) and the polishing stations **114**A-**114**D. The load cups **124**A, **124**B transport the substrates **108** between the carrier heads **120**A-**120**D and substrate exchangers **130** (e.g., individual exchangers **130**A, **130**B). A first substrate exchanger **130**A rotates in a first direction **132**A and a second exchanger **130**B rotates in a second direction **132**B, which may be opposite or the same as the first direction **132**A.
(13) The cleaning and drying section **104** includes a robot **136** that transfers the substrates **108** through the pass-through **110** to and from the substrate exchangers **130**A, **130**B at various access locations **172**A, **172**B. The robot **136** also transfers the substrates **108** between stations (not shown) in the cleaning and drying section **104** and the substrate exchangers **130**A, **130**B.
(14) FIGS. **2**A and **2**B depict a side view and a bottom view of the carrier head **120** according to some embodiments, which can be any of carrier heads **120**A-**120**D in FIG. **1**. The carrier head **120** includes a retaining ring **206** to retain the substrate **108** below a membrane **204**. The membrane **204** is a flexible, hydrophobic membrane **204**. The membrane outer perimeter **205** is surrounded by an inner perimeter **207** of the retaining ring **206**. A gap **216** is formed between the inner perimeter **207** of the retaining ring **206** and the outer perimeter **205** of the membrane **204**. In some embodiments, the gap **216** is about 0.5 mm to about 3 mm, such as about 1 mm to about 2 mm. The carrier head **120** includes one or more independently controllable pressurizable chambers (e.g., **202**A, **202**B, **202**C, **202**D) defined by the membrane **204**. Each of the pressurizable chambers have associated pressures (P.sub.A, P.sub.B, P.sub.C, and P.sub.D), and P.sub.R refers to the pressure exerted on the retaining ring **206** during processing. During processing, as the carrier head **120** rotates the substrate **108** while pressing it against the polishing pad (e.g., **116**A-**116**D), polishing slurry, debris and residue can accumulate on the substrate **108** edge, the substrate bevel areas, and other locations such as within the gap **216**.
(15) Without being bound by theory, it is believed that because the membrane **204** is hydrophobic,

the capillary and/or meniscus forces surrounding the outer perimeter **205** of the membrane **204** prevent conventional rinsing water, e.g., deionized (DI) water, from readily penetrating these gaps and features. The residue and particles can build up over time and be released during processing potentially causing scratches on the substrate **108**. One solution may be to rinse the membrane **204** with the membrane **204** facing upwards; however, this process is not conventionally used in the industry at least because it would affect throughput and cause water to bead up. A rinse process with the membrane **204** facing down is unable to wet the hydrophobic membrane surfaces and is limited in effectiveness for cleaning. It has been discovered that using a high pressure steam dislodges slurry residues and particles by using both kinetic and thermal energies. In some embodiments, the steam is effective for cleaning the gaps in the carrier head **120** both when the substrate **108** is not in place, and when the substrate **108** is retained in the carrier head **120**.
(16) FIG. **3**A depicts a top, plan view of the load cup **124** (e.g., **124**A or **124**B) according to an embodiment. The load cup **124** includes a substrate station **350** that has an annular shape. The substrate station **350** moves vertically to place substrates onto a blade **334** of the substrate exchanger **130** (e.g., **130**A, **130**B) and to remove the substrates **108** from the blade **334**. The blade **334** is rotatable to the access location **172**A, **172**B for loading and unloading of the substrate **108** by the robot **136** (FIG. **1**).
(17) The substrate station **350** includes notches (e.g., **352**A, **352**B, **352**C) to receive the blade **334**. The distal end **344** of the blade is received by notches **352**B and **352**C. The proximate end of the blade **334** is received by notch **352**A. The substrate **108** rests on raised features of the substrate station **350**. As the substrate station **350** moves in an upward direction and removes the substrate **108** from the blade **334**, the substrate **108** is positioned within a plurality of pins **354**, which create a pocket to center the substrate **108**.
(18) The load cup **124** includes a nebulizer **356** having a plurality of various nozzles (e.g., **358**A, **358**B, **358**C, **358**D) configured to spray fluids (e.g., deionized water) onto the blade **334**, a substrate **108** (not shown in FIG. **3**A) on the blade **334**, a substrate on the carrier head **120**, and/or a carrier head **120** (not shown in FIG. **3**A) located above the load cup **124**. The nebulizer **356** includes a set of first nozzles **358**A disposed around the outer portion of the nebulizer **356**, for example, to rinse the substrate **108**, and a set of second nozzles **358**B disposed in an array along a diameter of the nebulizer **356**, for example, to rinse the membrane **204** of the carrier head **120**, when respectively positioned over the load cup **124**. The nebulizer **356** includes a set of third nozzles **358**C on the outer portion of the nebulizer **356** configured to spray portions of the carrier head **120**, such as the gap **216** between the outer perimeter **205** of the membrane **204** and the inner perimeter **207** of the retaining ring **206**, when the carrier head **120** is positioned over the load cup **124**, with or without the substrate **108**. The third nozzles **358**C (e.g., spray nozzles) are also configured to spray an outer edge of the substrate **108**, while retained in the carrier head **120**, such as a gap between the outer edge of the substrate **108** and the inner perimeter **207** of the retaining ring **206**. The third nozzles **358**C are coupled to a rinse solution, such as deionized water at room temperature, such as about 10° C. to about 40° C. Each of the third nozzles **358**C are coupled to atomizers.
(19) The nebulizer **356** includes a set of fourth nozzles **358**D (e.g., energized fluid nozzles). Each of the fourth nozzles **358**D are disposed proximate to each third nozzle **358**C on an upper surface of the nebulizer **356** at an outer portion of the nebulizer **356**. In some embodiments, the energized fluid is deionized water (DIW), DIW and nitrogen, DIW and clean dry air (CDA), water ice particles and nitrogen, water ice particles and CDA, carbon dioxide ice, DIW energized with ultrasonic or megasonic generators, or combination(s) thereof. Without being bound by theory, it is believed that certain mixtures including ice particles can be used to bombard and dislodge debris within small voids and gaps. The energized fluid is gas phase fluid and/or a mixed phase fluid, such as vapor and/or steam. The temperature of the energized fluid, such as steam is about 80° C. to about 150° C., such as about 100° C. to about 120° C., such as a temperature at or above a

saturation temperature of the fluid. The pressure applied to energize the fluid is about 30 psi to about 140 psi, such as about 40 psi to about 50 psi. Other pressures and temperatures are also contemplated, such as for dry ice and other energized fluids.
(20) In some embodiments, the fluid is energized by pressurizing a fluid, acoustically energized (e.g., via acoustic cavitation), pneumatically assisted (e.g., using liquid mixed with a pressured gas), or combination(s) thereof. Other methods and combinations are also possible. Acoustic cavitation includes ultrasonically or megasonically energizing the fluid to dislodge residue and debris. Acoustically energizing fluid uses a piezoelectric transducer (PZT) operating in a frequency range from a lower ultrasonic range (e.g., about 20 KHz) to an upper megasonic range (e.g., about 2 MHz). Other frequency ranges can be used. The shape of a suitable acoustic energy source generator (e.g., a PZT) is rectangular. The acoustic source generator is coupled to the fourth nozzle **358**D.
(21) The fourth fluid nozzles **358**D are oriented upwards, perpendicular (e.g., about 90 degrees) to the upper surface of the nebulizer **356**. Other angles relative to the upper surface of the nebulizer **356** is also contemplated, such as about 45 degrees to about 100 degrees, in which 45 degrees is angle that oriented radially outward relative to the nebulizer **356**. Additionally each of the fourth fluid nozzles **358**D are configured to direct fluid in a flat fan jet (e.g., **360** shown in FIG. **3**B). FIG. **3**B depicts a schematic side view of a spray pattern of a flat fan jet **360** for the third nozzle **358**C and/or the fourth nozzle **358**D according to some embodiments. The flat fan jet is substantially parallel with a portion of an inner perimeter of the annular substrate station **350** and a jet angle α pivoting at a tip of the fourth fluid nozzle **358**D from a first edge to a second edge of the flat fan jet is about 30 degrees to about 50 degrees, such as about 40 degrees. In some embodiments, the nebulizer **356** includes about 1 to about 5 fourth fluid nozzles **358**D, such as 2, 3, or 4 fourth fluid nozzles **358**D. Each of the fourth fluid nozzles **358**D are equally spaced around the outer portion of the nebulizer **356**. In some embodiments, the nebulizer **356** includes about 1 to about 5 third nozzles **358**C, and each fourth fluid nozzle **358**D is disposed proximate to a respective third nozzle **358**C.
(22) FIG. **4** includes a flow diagram of a method for processing substrate. The method **400** includes, an operation **402**, directing energized fluid from a set of energized fluid nozzles (e.g., **358**D) of a load cup (e.g., **124**) at an edge of a substrate (e.g., **108**) disposed in a carrier head (e.g., **120**). The carrier head **120** includes the retaining ring **206** to retain the substrate **108** below the membrane **204** of the carrier head **120**. The substrate **108** edge is maintained at a temperature of about 60° C. to about 70° C., as measured from outside of the carrier head **120**. In some embodiments, after polishing the substrate **108** in the polishing section **102**, the substrate **108** is cooled to room temperature, such as about 20° C. to about 40° C. during rinsing and/or removing the substrate **108** from the polishing section **102**. It is believed that maintaining a low substrate **108** temperature, reduces a potential for corrosion during substrate transfer. The method described herein controls substrate temperature during cleaning, such as by localizing energized fluid to the substrate edge areas. In some embodiments, the energized fluid (e.g., steam) is simultaneously directed to the edge of the substrate **108** while spraying DI water from spray nozzles (e.g., **358**C) at the edge of the substrate **108**. In some embodiments, the DI water is sprayed immediately after the energized fluid is directed to the edge of the substrate **108**. The combined jet of the DI water and the energized fluid is controlled by adjusting the pressure of the energized fluid and the amount of water sprayed. In particular, a temperature of the combined jet is controlled to maintain the temperature of the substrate **108** below about 70° C. It is believed that maintaining a substrate **108** temperature below about 70° C. reduces undesirable effects on the substrate **108**. Residue and debris is dislodged from the edge of the substrate **108** without overheating the substrate and portions of the substrate carrier head **120** such as the retaining ring **206**. Managing temperature reduces the risk of material degradation of the substrate **108** and carrier head **120**. Spraying water also acts to rinse away the dislodged residue and debris. In some embodiments, the energized fluid

is directed to the substrate edge while the substrate **108** is in the carrier head **120** and rotating with the carrier head. The substrate **108** is secured to the carrier head **120** by application of a vacuum pressure to the membrane. The low clearance between the inner perimeter of the membrane and the substrate edge in combination with the hydrophobic properties of the membrane make it difficult to clean the substrate edge and bevel areas using conventional methods. The methods described herein provides an energized fluid having a fan jet shape and angle that is directed to the substrate edge while the substrate is secured to the carrier head **120** for enhanced cleaning.
(23) In operation **404**, the substrate **108** is unloaded from the carrier head **120**, such as by loading the substrate **108** onto a substrate exchanger (e.g., **130**A, **130**B) and rotating the substrate exchanger. After the substrate **108** is removed from the carrier head, the load cup **124** is returned to a position proximate the carrier head **120**, and energized fluid is directed to the empty carrier head **120** at the gap **216** formed between the outer perimeter **205** of the membrane **204** and the inner perimeter **207** of the retaining ring **206** in operation **406**. In some embodiments, DI water is sprayed immediately after the energized fluid to rinse away any dislodged residue and debris.
(24) Thus, the present disclosure relates to load cups that are configured with energized fluid nozzles which expel tunable energized fluid jets to clean the edges of a substrate disposed in a carrier head before the substrate is unloaded from the carrier head without overheating the substrate. The energized fluid jets expelled from the energized fluid nozzles have characteristics that are favorable for penetrating and effectively cleaning narrow gaps in the carrier head between a hydrophobic membrane and an inner perimeter of a retaining ring of the carrier head. Each of the energized fluid nozzles can be used in combination with respective fluid spray nozzles (e.g., deionized water spray nozzles) for improved control such as temperature control.

Claims

1. A load cup comprising: an annular substrate station configured to receive a substrate; and a nebulizer located within the load cup and surrounded by the annular substrate station, the nebulizer comprising: a set of energized fluid nozzles disposed on an upper surface of the nebulizer adjacent to an interface between the annular substrate station and the nebulizer, wherein the set of energized fluid nozzles are configured to release energized fluid at an upward angle relative to the upper surface, the upward angle being about 45 degrees to about 100 degrees relative to the upper surface, the energized fluid nozzles configured to direct a fluid in a flat fan jet, wherein a flat portion of the flat fan jet is substantially parallel with an inner perimeter of the annular substrate station; and a set of fluid nozzles disposed on the upper surface of the nebulizer, radially outward of the set of energized fluid nozzles, wherein each fluid nozzle of the set of fluid nozzles is disposed proximate to a respective energized fluid nozzle of the set of energized fluid nozzles.
2. The load cup of claim 1, wherein the set of fluid nozzles is configured to deliver a spray of fluid.
3. The load cup of claim 2, wherein the energized fluid is steam and the spray of fluid is deionized (DI) water.
4. The load cup of claim 2, wherein a jet angle pivoting at a tip of the energized fluid nozzle from a first edge to a second edge of the flat fan jet is about 30 degrees to about 50 degrees.
5. The load cup of claim 2, wherein each of the set of fluid nozzles are coupled to atomizers such that fluid from the fluid nozzles is in a mist form.
6. The load cup of claim 1, wherein the upward angle of the energized fluid is about 90 degrees relative to the upper surface of the nebulizer.
7. The load cup of claim 1, wherein the set of energized fluid nozzles include about 1 to about 5 energized fluid nozzles equally spaced around the upper surface of an outer portion of the nebulizer.
8. A chemical mechanical polishing system comprising: a carrier head comprising a retaining ring to retain a substrate below a membrane of the carrier head; and a load cup comprising: a set of

energized fluid nozzles disposed on an upper surface of an outer portion of the load cup, wherein the set of energized fluid nozzles are configured to direct energized fluid about 45 degrees to about 100 degrees relative to the upper surface and to a gap between an outer edge of the membrane and an inner perimeter of the retaining ring and are configured direct energized fluid in a flat fan jet, wherein a flat portion of the flat fan jet is substantially parallel with a portion of an inner perimeter of the retaining ring; and a set of fluid nozzles disposed on the upper surface of a nebulizer, radially outward of the set of energized fluid nozzles, wherein each fluid nozzle of the set of fluid nozzles is disposed proximate to a respective energized fluid nozzle of the set of energized fluid nozzles, each fluid nozzle directed about 45 degrees to about 100 degrees relative to the upper surface and towards the gap.
9. The chemical mechanical polishing system of claim 8, wherein a jet angle pivoting at a tip of each nozzle of the set of energized fluid nozzles from a first edge to a second edge of the flat fan jet is about 30 degrees to about 50 degrees.
10. The chemical mechanical polishing system of claim 8, further comprising atomizers coupled to each set of fluid nozzles.
11. The chemical mechanical polishing system of claim 8, wherein the set of energized fluid nozzles are disposed to such that the energized fluid is sprayed at about 90 degrees relative to an upper surface of the load cup.
12. The chemical mechanical polishing system of claim 8, wherein the membrane is hydrophobic and the gap formed between an outer perimeter of the membrane and the inner perimeter of the retaining ring is about 0.5 mm to about 2 mm.
13. A chemical mechanical polishing system comprising: a carrier head comprising a retaining ring to retain a substrate below a membrane of the carrier head; a load cup comprising: a first array of fluid nozzles disposed on an upper surface of an outer portion of the load cup, wherein the first array of fluid nozzles are directed about 45 degrees to about 100 degrees relative to the upper surface and towards a gap between an outer edge of the membrane and an inner perimeter of the retaining ring; a second array of fluid nozzles, each fluid nozzle of the second array of fluid nozzles disposed proximate and radially outward of a respective fluid nozzle of the first array of fluid nozzles, the second array of fluid nozzles being flat fan jet nozzles directed at the gap.
14. The chemical mechanical polishing system of claim 13, wherein the second array of fluid nozzles are arranged such that their spray is substantially parallel with a portion of an inner perimeter of the retaining ring.
15. The chemical mechanical polishing system of claim 13, wherein the flat fan jet of the second array of fluid nozzles is about 30 degrees to about 50 degrees.
16. The chemical mechanical polishing system of claim 13, further comprising a sonic generator coupled to the first array of nozzles.
17. The chemical mechanical polishing system of claim 13, wherein the membrane is a hydrophobic membrane.
18. The chemical mechanical polishing system of claim 13, wherein the load cup further comprises a nebulizer disposed radially inward of a movable substrate station.
19. The chemical mechanical polishing system of claim 18, wherein the first array of fluid nozzles and second array of fluid nozzles are disposed on the nebulizer.
20. The chemical mechanical polishing system of claim 19, further comprising a third array of fluid nozzles disposed in an array along a diameter of the nebulizer.